

1

2

#3D251F
cocap

#3DA2B7
teal

#ED6758
salmon

#FEC44C
citrus

#D9B37E
beige

#AA664A
copper

$$3 = \begin{cases} 0 & \text{if } \mathbf{C} - \mathbf{f} \oplus \mathbf{g} \geq 0 \\ -\infty & \text{else} \end{cases}$$

Deriving the (Discrete) Dual

Primal Problem: $\min_{\pi \in \mathbb{R}_+^{n \times m}} \langle \mathbf{C}, \pi \rangle \quad \text{s.t.} \quad \pi \mathbf{1} = \mathbf{a}, \pi^\top \mathbf{1} = \mathbf{b}$ “Lagrangian-ize constraints”

$$H(\mathbf{f}, \mathbf{g}) \triangleq \min_{\pi \in \mathbb{R}_+^{n \times m}} \langle \mathbf{C}, \pi \rangle + \mathbf{f}^\top (\mathbf{a} - \pi \mathbf{1}) + \mathbf{g}^\top (\mathbf{b} - \pi^\top \mathbf{1})$$

Let π_* be the optimal solution of the primal problem. We must have:

$$H(\mathbf{f}, \mathbf{g}) \leq \langle \mathbf{C}, \pi_* \rangle + \mathbf{f}^\top (\mathbf{a} - \pi_* \mathbf{1}) + \mathbf{g}^\top (\mathbf{b} - \pi_*^\top \mathbf{1}) = \langle \mathbf{C}, \pi_* \rangle$$

Now, rewrite:

$$H(\mathbf{f}, \mathbf{g}) = \underbrace{\mathbf{f}^\top \mathbf{a}}_1 + \underbrace{\mathbf{g}^\top \mathbf{b}}_2 + \underbrace{\min_{\pi \geq 0} \langle \pi, \mathbf{C} - (\mathbf{f} \oplus \mathbf{g}) \rangle}_3$$

Note: $3 = \begin{cases} 0 & \text{if } \mathbf{C} - \mathbf{f} \oplus \mathbf{g} \geq 0 \\ -\infty & \text{else} \end{cases}$

$$\max_{\mathbf{f}, \mathbf{g}} H(\mathbf{f}, \mathbf{g}) = \max_{\mathbf{f}, \mathbf{g}} \mathbf{f}^\top \mathbf{a} + \mathbf{g}^\top \mathbf{b} \quad \text{s.t.} \quad \mathbf{f} \oplus \mathbf{g} \leq \mathbf{C}$$

Un-lagrangianize constraints

Implications